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The TAs and I will not answer questions during the exam. Do not ask.

The exam is closed book, closed notes. No calculators, smart watches, smart phones, or other devices are allowed.

Write your answers entirely inside the provided boxes.

Points are earned only by what is written inside each box: write legibly, erase diligently.

All boxes have ample space for the correct answer.

Do supporting work on backs of pages.

I will neither give nor receive aid on this exam. [Please abide by this policy.]

Sign here

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DO NOT turn the page until told to start the rest of this exam. Thank you.

Check that this document contains all pages. If not, obtain a copy that does have all pages before starting to answer questions.

For multiple choice questions, answer using the letter of your choice.

Ensure that each of your answers is contained entirely within the appropriate box.

Write legibly.

Use the back of document pages for scratch work.

- ☐ 1. [Answer one of choices “A” and “B” instead of false or f or true or t. Thank you.]
Temporal locality means that if a machine instruction or a data item is accessed, then other instructions or data items nearby in the address space are likely to be accessed soon.
A The statement is false.
B The statement is true.
Answer: A. Describes spacial locality.
- ☐ 2. Given an LRU policy, the block to be replaced is the one in the set that has been resident in the cache for the longest time.
A The statement is false.
B The statement is true.
Answer: A.
- ☐ 3. There may exist multiple virtual memory address spaces at the same time in a running computer.
A The statement is false.
B The statement is true.
Answer: B.
- ☐ 4. A hardware lock can be used to enforce one program at a time write access to a shared variable.
A The statement is false.
B The statement is true.
Answer: B.
- ☐ 5. Communication time is always a minor overhead of MIMD computing.
A The statement is false.
B The statement is true.
Answer: A, false.
- ☐ 6. It is useful for software developers to track hardware technology changes because applications that are not feasible today due to hardware limitations may become feasible in the future.
A This statement is false.
B This statement is true
Answer: B.

- ☐ 7. Special purpose hardware is faster and uses less energy than does the same function implemented in software executed by a general purpose CPU.
A The statement is false.
B The statement is true.
Answer: B, true.
- ☐ 8. CAM must contain a power of 2 number of storage locations.
A The statement is false.
B The statement is true.
Answer: A, false.
- ☐ 9. The greater the number of items stored in a content addressable memory, the greater the number of search steps needed to determine if a key is present in that CAM.
A The statement is false.
B The statement is true.
Answer: A.
- ☐ 10. If any one of the inputs to a binary full adder circuit is changed to the other logic level, then after the worst case propagation delay for the full adder
A the Carry_out output has changed.
B the Carry_out output equals 0.
C the Sum output has changed.
D the Sum output equals 1.
E None of the other answers is correct.
Answer: C.
- ☐ 11. [Write your answer as a decimal integer only. No word(s).]
How many Mebibits is 2^{23} bytes?
Answer: 64.
- ☐ 12. Which of these assembly language instructions replaces the default next instruction pointer with a pointer that has been stored?
A Branch
B Return
C Jump
D All of the above
E None of the above
Answer: B.
- ☐ 13. A multiplexer has a total of 17 input and output buses. How many bits wide is its address bus?
Answer: 4.
- ☐ 14. Which circuit implements the pointing function?
A binary counter
B decoder

- C multiplexer
- D register
- E None of the other answers is correct.

Answer: B.

- ☐ 15. The bit string 01011010 can be written as
- A 0x1122
 - B 0x01011010
 - C 0x5a
 - D 0xA5
 - E The question makes no sense because 8-bit strings have no inherent meaning.

Answer: C. From text chapter 3, section 7.

- ☐ 16. Which is a criterion used to guide the design of ASCII code choices?
- A Making full use of one byte of memory.
 - B Use of hexadecimal encoding.
 - C The use of the NULL character as a string terminator.
 - D Ease of machine use.
 - E The question makes no sense because ASCII was not designed, it is a standard.

Answer: D.

- ☐ 17. Because a processor circuit performs the fetch-execute cycle when it is supplied with electrical power, software must ensure that there
- A is electrical power.
 - B are occasional rest periods for the processor circuit to cool off.
 - C is an operating system present in memory.
 - D is always a next instruction.
 - E is an unlimited quantity of data to process.

Answer: D.

- ☐ 18. Write the result of sign-extending the bit string 0x7f to 16 bits. Use hexadecimal notation for your answer.

Answer: 0x007f

- ☐ 19. Control flow in a machine language program is
- A managed by computation of a pointer.
 - B performed by the ALU.
 - C handled by the register unit.
 - D unimportant until after a program is compiled using the computer that will then execute it.
 - E related to the size of the registers in the register unit.

Answer: A.

- ☐ 20. Choose the best way to complete the following sentence. An operand value that is known at compile time is
- A stored in a register as a bit string.

- B found at the output of data memory.
- C stored in a machine instruction if its magnitude is small.
- D a way to increase the speed of compilation.
- E not possible.

Answer: C.

- ☐ 21. A 32-bit machine instruction can be written
- A as a 4-byte bit string.
 - B using 4 hexadecimal digits.
 - C as an 8-byte string.
 - D as an 8-byte string.
 - E None of the other answers is correct.

Answer: A.

- ☐ 22. A dirty bit in L1 data cache is set when
- A it is time to enforce Rule 1 of the memory hierarchy.
 - B a new block is brought into the L1 data cache.
 - C accessing a given cache block for the second time.
 - D the processor executes a store operation.
 - E None of the other answers is correct.

Answer: D

- ☐ 23. This data structure, maintained locally in each Processing Element of an SIMD computer, allows each PE to correctly execute or ignore computational instructions that are contained in a while statement or an if-else statement.

Answer: enable stack.

- ☐ 24. Which one quantifies the overall performance improvement of a system when a subset of the system is improved?
- A Moore's Law
 - B The CPU time equation
 - C Flynn's MIMD category
 - D Amdahl's Law
 - E None of the other answers is correct.

Answer: D.

- ☐ 25. [Answer in the form of a decimal integer.]
How many comparators are there in a 64-block fully associative cache?

Answer: 64 or equivalent.

- ☐ 26. A 4-way set associative cache has a total of 1024 blocks. Each block is 16 bytes. Assume that memory is byte addressed and that the cache is accessed using a 32-bit memory address. Bit position 31 of a memory address is the MSB and position 0 is the least significant bit. Let the notation "[8-3]" mean the address bits in positions 8 through 3, inclusive.

Which bits of a memory address are used as the set number field?

Answer: [11-4].

27. What is the 4-bit binary sign magnitude representation for decimal integer (-2)?

Answer: 1010.

28. A program instruction sequence has no branches in, except to the first instruction, and only one branch out, which is the last instruction of the sequence. What is the name for this program structure?

Answer: basic block.

29. A computer has 10 processors that work together as a team to solve large problems. What is the Flynn parallelism category that describes this computer?

Answer: MIMD

30. What is reported as the result of a response-type SPEC benchmark?

Answer: elapsed time or wall clock time

31. The company AMD claims for their Zen3 processor architecture that for a set of benchmark programs the geometric mean of the number of instructions issued to the processor backend is improved by 19% with respect to the same benchmarks executed by the AMD Zen2 architecture. This statement means that, overall, the number of instructions issued by the Zen3 design during each clock cycle when compared to the Zen2 design is

- A increased.
- B decreased.
- C the same.
- D not comparable.
- E None of the other answers is correct.

Answer: A.

32. [Answer “yes” or “no”.]

The equation for average memory access time, AMAT, is

$$\text{AMAT} = \text{Hit_rate} \times \text{Hit_time} + (1 - \text{Hit_rate}) \times \text{Miss_time} .$$

Designers of a processor decide to increase the capacity of the L1 data cache. This causes hit time for this cache to increase due to the larger SRAM array necessary to increase capacity. The designers change nothing else about the memory hierarchy of this processor.

Because Hit_time has increased, must Average_Memory_Access_Time increase when executing a given set of benchmark programs?

Answer: no.

33. Amdahl's Law models overall speedup, S_{overall} , due to an enhancement that provides a speedup, S_e , and that can be used a fraction, f_e , of the original (unenhanced) execution time. Amdahl's Law is

$$S_{\text{overall}} = \frac{1}{(1-f_e) + \left(\frac{f_e}{S_e}\right)}.$$

The equation for average memory access time, AMAT, is

$$\text{AMAT} = \text{Hit_rate} \times \text{Hit_time} + (1-\text{Hit_rate}) \times \text{Miss_time}.$$

Which variable in the AMAT equation corresponds to f_e in Amdahl's Law?

Answer: Hit_rate.

34. The speedup provided by an enhancement is H. What is the upper bound on the overall speedup possible for a system due to this enhancement?

Answer: H.

35. [Express your answer as C language keyword(s) only.]
Consider the following assembly language level code snippet.
here: instruction to compute a condition
branch to there if condition is false
instructions to compute statement body
jump to here

there: instruction for next statement

What high-level language statement is expressed by the assembly language program?

Answer: while.

36. [Express your answer in the form of a decimal integer.]
Consider the following assembly language code snippet.
- ```
LOAD R4, R1+0 ;R4 ← value at address R1 + 0
ADD R6, R4, R5 ;R6 ← R4 + R5
STORE R6, R1+0 ;Store R6 at address R1 + 0
ADDI R1, R1, #4 ;R1 ← R1 + 4
```

An anti-dependence is when a location holds a data value that the program reads and then later writes. How many anti-dependences appear in this code snippet?

Answer: 3.

From LOAD reading R1 to ADDI writing R1.

From STORE reading R1 to ADDI writing R1.

From LOAD reading at memory address R1+0 to STORE writing to memory address R1+0, where the content of R1 did not change between the execution of the LOAD and the STORE.

37. [Express your answer in the form of a decimal integer.]  
Consider the following assembly language code snippet.
- ```
LOAD R4, R1+0      ;R4 ← value at address R1 + 0
ADD R6, R4, R5      ;R6 ← R4 + R5
STORE R6, R1+0      ;Store R6 at address R1 + 0
```

ADDI R1, R1, #4 ; R1 $\leftarrow$ R1 + 4

An output dependence is when a program writes a result to a location and later writes to that location again. How many output dependences appear in this code snippet?

Answer: 0.

Locations written to by each instruction are, in program order, R4, R6, Data_Memory[R1+0], and R1. No location is written to more than once in the snippet. Thus, there are zero output dependences.

38. [Express your answer as a decimal integer.]

A 64-bit computer uses 4 Kibibyte pages and 64-bit virtual addresses. This computer divides the virtual page number into 4 equally-size bit strings to then store page tables as 4-level trees. A pointer to a child node in a page table tree is stored in 8 bytes. A virtual page table entry is stored using 8 bytes.

Assume that some program uses a total of 7 virtual pages in the virtual memory address space. For this program, the best-case total size of the tree-structured page table in bytes is a power of 2. What is this exponent?

Answer: 18.

The best case is when each of the virtual pages used maps to the same leaf node. Thus, there will be 3 ancestor nodes, including the root and one leaf node for a total of 4 nodes in the tree-structured page table. Each tree node requires 2^{13} entries at 2^3 bytes each. Thus, the total size in bytes is $2^2 \times 2^{13} \times 2^3$ bytes = 2^{18} bytes.

39. The target of a particular assembly language branch instruction is located closer to the first instruction of the assembly program than is the branch instruction. When predicting this branch instruction what is the best fixed branch prediction to use?

Answer: predict taken or just “taken”.

40. The execution trace of a for loop of N iterations includes execution of N branch instructions. Now the loop is unrolled k times where k divides N. The execution trace of the unrolled loop includes just 20% as many branch instructions as without unrolling. What is the value of k?

Answer: 5.

41. You are actively editing a source code file on a computer with a memory hierarchy that includes an L1 cache. You save your work in progress to non-volatile storage, perhaps by making a commit to GitHub. You have just performed which cache write policy?

- A Write through
- B Write back
- C Saving editing work in progress does not involve a cache write action.
- D None of the other answers is correct.

Answer: A.

42. Which part of a virtual address is not stored in a single-level page table?
A virtual page number

- B** offset
- C** Each of the above answers names a part of a virtual address not stored in a single-level page table.
- D** None of the above answers is correct.
- E** There is insufficient information given to answer the question asked.

Answer: C.

43. A first-generation processor has a register unit constructed with 32-bit registers. The next generation of the processor replaces each 32-bit register in the register unit with a 64-bit register.

Consider a machine instruction that sources both operands from the register unit and stores their sum in the register unit. How many more bits does the next-generation processor machine instruction require to specify three registers than the first-generation machine instruction?

[State your answer as a decimal integer. Do not include the word “bits”. Thank you.]

Answer: 0.

44. This digital logic circuit is used to ignore all ALU function unit outputs except the one specified by the opcode.

Answer: multiplexer.

45. To execute two machine instructions at the same time requires an absence of what between the two instructions?

Answer: dependence or dependency.